

Orwell-Vernon estuary

Shoreline length: 72 km



Biophysical Environment

Orwell-Vernon estuary contains two tidal estuaries with multiple barrier spits and coastal infrastructure. Located on the east end of PEI, with a majority of low plains and sloped armour.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
53.3	10.4	18.0	1.1	14.4	1.8	0	1.1

Communities, Heritage, Infrastructure

The Orwell-Vernon estuary falls entirely the Town of Three Rivers and includes both Georgetown and Montague. Municipal planning is in place. No specific flood elevations or vertical setbacks established and specific erosion-based setbacks, possibility of reduced setbacks with armouring. Hazards are assessed at the subdivision stage.

Key infrastructure primarily exists in the communities of Georgetown and Montague. Georgetown includes the Georgetown Elementary School, Georgetown Volunteer Fire Department, Holland Collage, Kings Playhouse, the seafood wholesaler; Seafood 2000 Ltd., Georgetown Waterfront Boardwalk, A.A. MacDonald Memorial Gardens, and the Georgetown Sewage Collection and Treatment Corporation.

Key infrastructure in Montague includes Kings County Memorial Hospital, Montague Heath Centre, Montague Fire Department, Montague Consolidated School, Montague Regional High School, North Lake Fisheries – Montague Bay Foods, Awesome Adventures Fisheries, four gas stations, and Three Rivers Splash Pad.

Orwell-Vernon estuary also contains the Roma at Three Rivers National Historic Site reflecting the areas historical and cultural significance.

Parks, Protected Areas, Habitats

Orwell-Vernon estuary includes Brudenell Courts (tennis), Brudenell Provincial Park, West Street Parks (playground), Georgetown tennis courts, Georgetown elementary playground and gymnasium, AA MacDonald memorial park (playground), Brudenell River Golf course; Spruce Street Park (playground), Montague Waterfront Park (playground), Clay Park (playground), Montague consolidated field (soccer), Knox Dam (walking trail), Jean-Pierre Roma at Three Rivers National historic site, and the Georgetown Waterfront Boardwalk.

There are no species at risk in the Orwell-Vernon estuary.

% Shore distance with species at Risk

Endangered	Threatened
0	0

Planning and Policy Framework

Land use and development is managed under the Town of Three River's planning rules. Development on the coast is informed by the Environmental Protection Act, including the 15m buffer, with no additional building setbacks aside from the buffer. Three River's planning documents include specific guidance permitting armouring, but such rules would be secondary to any provincial standards. The plan includes some policies regarding expansion of buildings in sea level rise overlay. There are no specific flood elevations or vertical setbacks established but the official plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage. Local land use constraints and considerations include several small lot subdivisions (over 100 small lots) serviced by unpaved roads, often perpendicular to the shore, some at FR, including Newport and Seal River, as well as economic drivers such as Industrial farm operation + wharf, Water Street, Georgetown, Georgetown wharf and marina, Georgetown pier, Seafood 2000 Ltd fish plant and wharf at Georgetown, Georgetown West Street beach, Brudenell Canoe Kayak Club, Brudenell River Golf Course, Rodd Brudenell Resort, Montague Waterfront Marina, Montague fishing wharf, Lower Montague wharf.

Community Feedback

**To be updated following Fall 2026 engagement.*



Coastal Flood and Erosion Risk

Infrastructure along low plain and slope armour shorelines are vulnerable to increasing flood hazards. Erosion hazard is limited due to the sheltered orientation of the area. Erosion hazard is primarily seen at Poxy Island an exposed sand spit.

The area includes a number of potentially at risk roads and facilities, including Ayrton Ln; Brendons Bl; Brudenell Island Bl; Cest La Vie Ln; Conor Rd; Cook Cove Rd; Deedles Ln; Delodder Rd; Dewars Rd; Dewars Rd; East Royalty Rd; Grafton St; Harbourview Dr; High St; Kent St; Louis Wright Ln; Lower Montague Rd; Mair Rd; Mill St; Old Anchor Ln; Parkers Point Ln; Patrick St; Riverside Dr; Roma Point Rd; Shaws Ln; South Montague Rd - Rte 353; Sparrows Rd - Rte 320; Spring Water Ln; St Andrews Point Rd; Station St; Thornton Rd; Valleyfield Rd - Rte 326; Valleyfield Rd - Rte 326; Water St; West St Wight Birch Cr; and Wood Ln.

Average number of at-risk buildings per km of shoreline

	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)
Flood risk based on SLR and 100-year event	11	15	20	27
Erosion risk (shoreline change year) or salt marsh migration (SLR)	N/A	15	23	48

Sediment transport - The estimated net sediment transport rate averaged across the SMU is low. Sediment transport rate potential is highest along Burnt Point and Poxy Island.

SMP Strategies

The primary strategy for **Orwell-Vernon estuary** is Resist. This does not mean armouring the shoreline is the only strategy; it indicates that resist is one option to protect at risk infrastructure.

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	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)
Results of Decision Tool (%)			
No Active Intervention	57	TBD	TBD
Managed Realignment	11	TBD	TBD
Resist	32	TBD	TBD
Proposed Overall SMU Strategy	NAI except for special PUs	TBD	TBD
Rationale / comments / issues	E.g., Infrastructure along low plain and slope armour shorelines are vulnerable to increasing flood hazards.	E.g., Infrastructure is increasingly at risk due to SLR.	E.g., Infrastructure is increasingly at risk due to SLR.
Triggers for pivoting to next epoch	E.g., sea level rise, level of and cost of, flood damages.	E.g., sea level rise, level of and cost of, flood damages.	E.g., sea level rise, level of and cost of, flood damages.
Local Policy Units, Special Cases			
Georgetown and Montague	Resist - The core area of Georgetown and Montague may warrant special attention.	The core area of Georgetown and Montague may warrant special attention.	The core area of Georgetown and Montague may warrant special attention.
Legacy Constraints	Large estuary with several small lot subdivisions (over 100 small lots) serviced by unpaved roads, often perpendicular to the shore, unpaved roads at burnt point	Large estuary with several small lot subdivisions (over 100 small lots) serviced by unpaved roads, often perpendicular to the shore, unpaved roads at burnt point	Large estuary with several small lot subdivisions (over 100 small lots) serviced by unpaved roads, often perpendicular to the shore, unpaved roads at burnt point

Commented [VL1]: Check this with new Clenmar numbers

Action Plan

**SMU specific action plan to be determined and summarized here following the UK model. Example strategies provided in the table below.*

Action	Timeline for Implementation	Responsible Authority
Resist - Existing defences are expected to remain in place with limited maintenance until a new set back defence is built. Subject to maintenance and replacement of existing defences. Existing defences are expected to come to the end of their serviceable life in the medium or long term.	TBD	TBD
Managed realignment - Construction of new, realigned defences. Planning to use appropriate development controls and conditions to help manage the risk to businesses and residents and consider long term realignment. Existing defences are expected to remain in place with limited maintenance until a new set back defence is built	TBD	TBD
No active intervention – Erosion will lead to realignment of the shore, but high ground limits flood risk in most areas. There are no built defences in this policy unit but there is limited flood risk. Some areas may be at risk from high tides and surges. Planning to use appropriate development controls and conditions to help manage the risk to businesses and residents.	TBD	TBD

**Adaptation Pathways to be developed for SMUs that require guidance on shifting from on strategy to another.*

Maintain natural defences in the area fronting the foreshore reserve;
 maintain/rehabilitate dunes (native planting) and manage access
 Soft engineering – enhance the dune by setting back into the
 reserve and native planting
 Change planning practices relating to hazard affected properties –
 plan for retreat
 Retreat hazard affected properties and assets



• trend outside the normal erosion and accretion cycle has been demonstrated
 • dune varies along the length of the beach

